

In [2]:

```
pip install pandas
```

Requirement already satisfied: pandas in E:\New folder\Lib\site-packages (3.0.3)  
Requirement already satisfied: numpy>=2.3.3 in E:\New folder\Lib\site-packages (from pandas) (2.4.6)  
Requirement already satisfied: python-dateutil>=2.8.2 in E:\New folder\Lib\site-packages (from pandas) (2.9.0.post0)  
Requirement already satisfied: tzdata in E:\New folder\Lib\site-packages (from pandas) (2026.2)  
Requirement already satisfied: six>=1.5 in E:\New folder\Lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)  
Note: you may need to restart the kernel to use updated packages.

In [4]:

```
import pandas as pd
```

In [5]:

```
data=pd.read_csv(r"C:\Users\vv519\Downloads\trends.csv")
```

In [6]:

```
data
```

Out[6]:

|       | location | year | category        | rank | query               |
|-------|----------|------|-----------------|------|---------------------|
| 0     | Global   | 2001 | Consumer Brands | 1    | Nokia               |
| 1     | Global   | 2001 | Consumer Brands | 2    | Sony                |
| 2     | Global   | 2001 | Consumer Brands | 3    | BMW                 |
| 3     | Global   | 2001 | Consumer Brands | 4    | Palm                |
| 4     | Global   | 2001 | Consumer Brands | 5    | Adobe               |
| ...   | ...      | ...  | ...             | ...  | ...                 |
| 26950 | Vietnam  | 2020 | Là Gì?          | 1    | Virus Corona là gì  |
| 26951 | Vietnam  | 2020 | Là Gì?          | 2    | Miền thị thực là gì |
| 26952 | Vietnam  | 2020 | Là Gì?          | 3    | Đầu cắt moi là gì   |
| 26953 | Vietnam  | 2020 | Là Gì?          | 4    | Bệnh bạch hầu là gì |
| 26954 | Vietnam  | 2020 | Là Gì?          | 5    | Đông Lào là gì      |

26955 rows × 5 columns

In [17]:

```
data.shape
```

Out[17]:

(26955, 5)

In [18]:

```
data.head()
```

Out[18]:

|   | location | year | category        | rank | query |
|---|----------|------|-----------------|------|-------|
| 0 | Global   | 2001 | Consumer Brands | 1    | Nokia |
| 1 | Global   | 2001 | Consumer Brands | 2    | Sony  |
| 2 | Global   | 2001 | Consumer Brands | 3    | BMW   |
| 3 | Global   | 2001 | Consumer Brands | 4    | Palm  |
| 4 | Global   | 2001 | Consumer Brands | 5    | Adobe |

In [19]:

```
data.tail()
```

Out[19]:

|       | location | year | category | rank | query               |
|-------|----------|------|----------|------|---------------------|
| 26950 | Vietnam  | 2020 | Là Gì?   | 1    | Virus Corona là gì  |
| 26951 | Vietnam  | 2020 | Là Gì?   | 2    | Miền thị thực là gì |
| 26952 | Vietnam  | 2020 | Là Gì?   | 3    | Đầu cắt moi là gì   |
| 26953 | Vietnam  | 2020 | Là Gì?   | 4    | Bệnh bạch hầu là gì |
| 26954 | Vietnam  | 2020 | Là Gì?   | 5    | Đông Lào là gì      |

In [14]:

```
data.describe()
```

Out[14]:

|       | year         | rank         |
|-------|--------------|--------------|
| count | 26955.000000 | 26955.000000 |
| mean  | 2015.243369  | 3.000000     |
| std   | 3.564683     | 1.41424      |
| min   | 2001.000000  | 1.00000      |
| 25%   | 2013.000000  | 2.00000      |
| 50%   | 2016.000000  | 3.00000      |
| 75%   | 2018.000000  | 4.00000      |
| max   | 2020.000000  | 5.00000      |

In [15]:

```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 26955 entries, 0 to 26954
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   location    26955 non-null  str
1   year        26955 non-null  int64
2   category    26955 non-null  str
3   rank        26955 non-null  int64
```

```
4 query      26955 non-null str  
dtypes: int64(2), str(3)  
memory usage: 2.0 MB
```

```
In [ ]:
```